Reducibility 01

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Reducibility

- **Reducibility** is a way to convert a problem A into another problem B where
  - a solution to the problem B can be used to solve problem A
- Problems in our discussion are membership problems
- Consider two sets $A \subseteq \mathcal{X}$ and $B \subseteq \mathcal{Y}$ and a function $f : \mathcal{X} \to \mathcal{Y}$ such that

$$w \in A \leftrightarrow f(w) \in B$$

- Recall that

$$\begin{aligned} P \leftrightarrow Q &\equiv (P \to Q) \wedge (Q \to P) \\ &= (\neg P \vee Q) \wedge (\neg Q \vee P) \\ &= (Q \vee \neg P) \wedge (P \vee \neg Q) \\ &= (\neg Q \to \neg P) \wedge (\neg P \to \neg Q) \end{aligned}$$

- If the above property holds, we have:
  - If $w \in A$, $f(w) \in B \rightsquigarrow$ If $f(w) \notin B$, $w \notin A$
  - If $f(w) \in B$, $w \in A \rightsquigarrow$ If $w \notin A$, $f(w) \notin B$

Reducibility

- Again, consider two sets $A \subseteq \mathcal{X}$ and $B \subseteq \mathcal{Y}$ and a function $f : \mathcal{X} \to \mathcal{Y}$ with the following property:

$$w \in A \leftrightarrow f(w) \in B$$

- Problem #1: Is a specific element in the set A
- Problem #2: Is a specific element in the set B
- Reducibility:
  - Given an element $w \in \mathcal{X}$, suppose no idea whether $w \in A$
  - However, given an element $s \in \mathcal{Y}$, we know whether $s \in B$
  - Suppose $f()$ exists and the above property hold
  - Convert w to $f(w)$ and check whether $f(w) \in B$
    - If $f(w) \in B$, $w \in A$
    - If $f(w) \notin B$, $w \notin A$
- Solution to problem #2 can be used to solve problem #1

Example

- Suppose we have a sport car and we need to know whether it can go over 200 MPH
- Unfortunately, we cannot drive our car that fast in Pittsburgh
  - Way too many potholes
- Luckily, we have a psychic lady who has a very special ability
  - She can tell whether an object can explode into multiple pieces
- So, let's put a bomb into our sport car
  - Put the trigger at the speedometer at 200 MPH mark
- Take the car to the psychic lady:
  - If she says that it can explode $\rightsquigarrow$ our sport car can go over 200 MPH
  - If she says that it cannot explode $\rightsquigarrow$ our sport car cannot go over 200 MPH

Example

- Let's compare the previous example with the concept of reducibility
- Let w be our sport car
- Let $\mathcal{X}$ be the set of cars
- The set $A \subseteq \mathcal{X}$ can be

$$A = \{c \mid c \text{ is a sport car that can go over 200 MPH}\}$$

- Problem #1: Is our sport car in the set A? (Is $w \in A$?)
  - We want to know the answer to this problem
- Let $\mathcal{Y}$ be the set of objects
- The set $B \subseteq \mathcal{Y}$ can be

$$B = \{x \mid x \text{ is an object that can explode}\}$$

- Problem #2: Is an object s can explode? (Is $s \in B$?)
  - The psychic lady knows how to answer this problem

Example

- The function $f : \mathcal{X} \to \mathcal{Y}$ converts a car into an object by strapping a bomb to the car and wire the trigger into the speedometer
- The function $f()$ converts an instance of the problem #1 into an instance of the problem #2 such that

$$w \in A \leftrightarrow f(w) \in B$$

- Recall that we put a bomb into our car (w) and wire the trigger to speedometer
  - We convert w into $f(w)$
  - We convert an instance of the problem #1 into and instance of the problem #2

Example

- Then we ask the psychic lady whether $f(w) \in B$
- The answer from the psychic lady can be use to answer our first problem because of the above property
  - If the psychic lady says $f(w) \in B \rightsquigarrow w \in A$
    - Our sport car can go over 200 MPH
  - If the psychic lady says $f(w) \notin B \rightsquigarrow w \notin A$
    - Our sport car cannot go over 200 MPH

Reducibility

A is **reducible** to B if a solution to the problem B can be used to solve problem A.

Reducibility

- Suppose B is a language
  - What does it mean by the psychic lady know whether an object is in B?
  - B is decidable
  - B is decide by the psychic lady
- Suppose A is a language
  - What happen if $f()$ exists and the following property hold?

$$x \in A \leftrightarrow f(x) \in B$$

  - A is decidable by the following TM

$M =$ "On input x where x is a string:

1. Convert x to $f(x)$
2. Run the psychic lady on input $f(x)$
3. If the psychic lady accept $f(x)$, $accept$. Otherwise, $reject$."

Reducibility

If B is decidable and $f : \Sigma^* \to \Sigma^*$ exists such that

$$x \in A \leftrightarrow f(x) \in B$$

A is decidable.

- Again, let's apply some boolean algebra:
  - $P \equiv B$ is decidable
  - $Q \equiv f : \Sigma^* \to \Sigma^*$ exists such that ...
  - $R \equiv A$ is decidable
- According to the boolean algebra, we have

$$(P \wedge Q) \to R \equiv (\neg R \wedge Q) \to \neg P$$

If A is undecidable and $f : \Sigma^* \to \Sigma^*$ exists such that

$$x \in A \leftrightarrow f(x) \in B$$

B is undecidable.

Undecidability

- How to prove that a language A is undecidable?
  - Use the fact that A_{TM} is undecidable
- But how?
  - Imagine if you can prove that

    If A is decidable, A_{TM} is decidable

  - The above statement is in the form of $P \rightarrow Q$
  - Since $P \rightarrow Q \equiv \neg Q \rightarrow \neg P$, we have

    If A_{TM} is undecidable, A is undecidable

  - This allows you to conclude that A is undecidable because A_{TM} is undecidable

Use A_{TM}

- To show that a language A is undecidable (proof by contradiction)
  1. Assume that A is decidable
  2. Show that if A is decidable, A_{TM} is also decidable
  3. Since A_{TM} is decidable — contradiction
  4. Therefore, A is undecidable
- But why all the sudden A_{TM} is decidable?
  - By assuming that a language A is decidable, it means there exists a TM R that decides A
    - R is a decider for A
    - $L(R) = A$
    - R always halts
    - TM R can tell you whether a specific element is in A or not
    - If $w \in A$, R accepts w
    - If $w \notin A$, R rejects w
  - This TM R will be our helper to help us decide A_{TM}